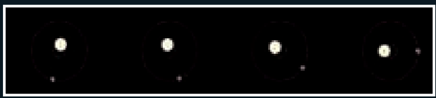


A star pirouetting about the green circle due to a planet being present in orbit

rimeters which are capable of differentiating polarised beams of light. However, such instruments require as little visual aberration as possible and are used on space telescopes, not terrestrial ones.

Interferometers

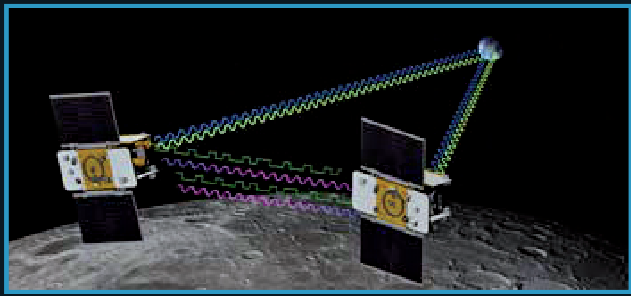
Interferometry is the process of combining the light from two telescopes to get a much better picture. So multiple small telescopes can be used to collect data which can be merged to produce data that would otherwise require a really huge telescope. A trick that astronomers use with this method is called "Null Interferometry". Light travels in waves and when two waves are brought together by aligning the "troughs" together, the signal is boosted to give a much better picture. Aligning the trough of the wave to the peak of the waves will cancel that particular source of light. Thus, the star's light is nullified to view the planets better.

Colour of a planet

A more recent technique discovered by astronomers has helped figure out the colour of light reflected off from a planet's surface. When a planet orbits a star, it moves away from us for one-half of its revolution and moves closer to us during the other. The light observed from the planet is either red/blue shifted according to the movement of the planet. By continuing this shifting, astronomers were able to determine the light spectrum reflected by the planet by honing onto it. This not only confirmed the existence of the planet but also gave them information about it.

Conclusion

In a young galaxy as ours there's much to be discovered and studied, after all resources are limited and the only way to sustain our growth would be to venture into the cosmos, expanding our horizons. Learning more about the cosmos is our first step in this direction. d



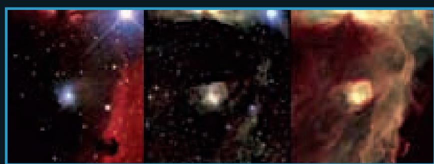
Satellites GRAIL-A and GRAIL-B studying the Moon's gravitational field

GRAIL

Gravity Recovery And Interior Laboratory is part of NASA's Discovery Space program which launched two probes recently (these will orbit the moon and study its gravitational field). When a planet cools down, the dense materials move towards the centre of the planet. There's a constant change in the gravitational field when this happens. Studying this will help expand our understanding of the evolution of the moon and the knowledge gained will be applied to further our understanding of planet formation as well.

Red shift

Redshift is a phenomenon in which light from a body is increased in wavelength. This happens when an object from which the light is being emitted is in motion. It's like the doppler effect which can be experienced when a sound emitting object goes past you. The sound you hear prior to the object passing you is of a higher frequency than the sound you hear once the object goes past you. This is because there's compression of the sound waves in the direction towards which the object is moving while expansion occurs right behind it. The same is applicable to light which also travels as waves. The compression results in the light entering the blue spectrum, this is termed as "blueshift", whereas expansion results in "redshifting". So by measuring the redshift of an object we can learn much about the planet's motion.



Normal, near-infrared and mid-infrared images of a Nebula viewed from the Hubble Space Telescope. Observe as certain elements are only visible when viewed from a particular infrared range.